

DATA SCIENCE INTERNSHIP REPORT

Submitted To

Cognifyz Technologies

Submitted By :-

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Introduction

This internship focused on practical Data Science techniques using Python. The project involved exploring a restaurant dataset, performing descriptive analysis, geospatial analysis, feature engineering, and data preprocessing. The tasks enhanced my knowledge of Python, data analysis, and visualization.

Dataset Description :

- Dataset Name: Restaurant Dataset
- Total Rows: 9551
- Total Columns: 21
- File Type: CSV

Tools Used:

- Python
- Pandas
- Matplotlib
- VS Code

Level 1 – Task 1

Data Exploration & Preprocessing

Objective

To explore the dataset and identify missing values, data types, and the distribution of the target variable.

Work Done

Explored dataset structure.

- Checked missing values.
- Identified data types.
- Analyzed Aggregate Rating distribution.

Result

The dataset contains **9551 rows** and **21 columns**. Only a few missing values were found in the **Cuisines** column. The target variable showed an imbalanced distribution.

Level 1 – Task 2

Descriptive Analysis

Objective

Perform statistical analysis on numerical and categorical columns.

Work Done

- Calculated Mean
- Median
- Standard Deviation
- Identified Top Cities
- Identified Top Cuisines

Result

The dataset revealed that **New Delhi** has the highest number of restaurants. Popular cuisines include **North Indian**, **Mughlai**, **Fast Food**, and **Cafe**.

Level 1 – Task 3

Geospatial Analysis

Objective

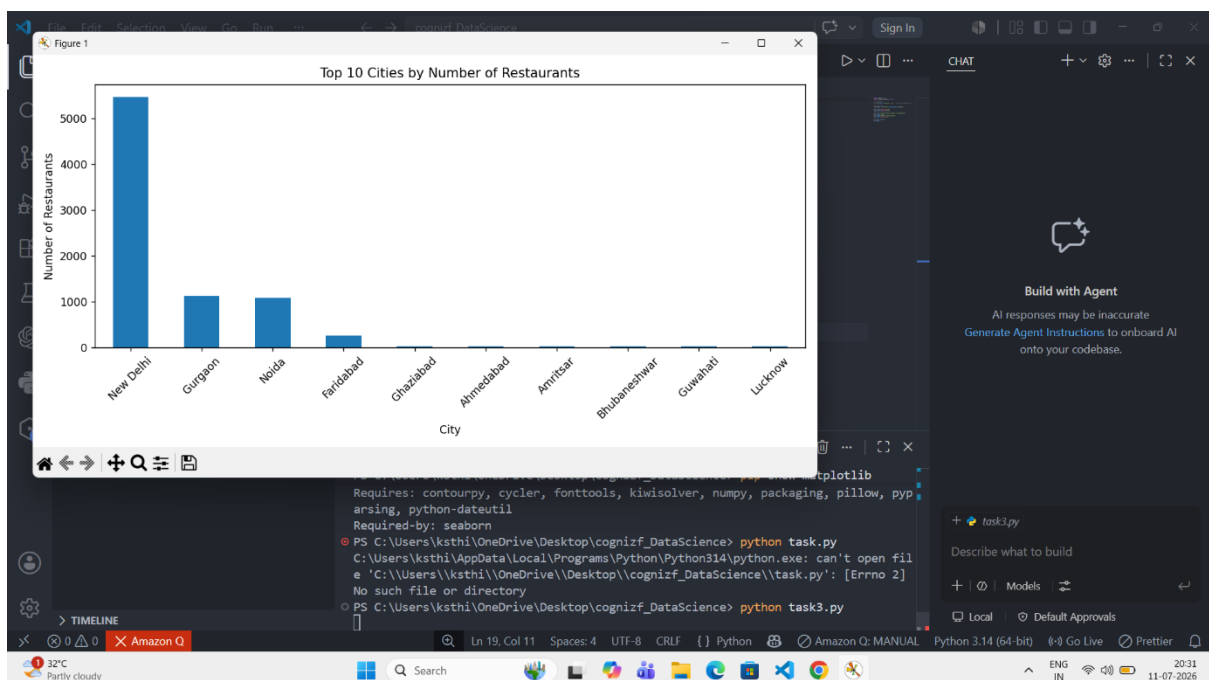
Analyze restaurant distribution based on location.

Work Done

- Created Bar Chart
- Compared Top 10 Cities

Result

New Delhi has the highest restaurant concentration followed by Gurgaon and Noida.



Level 2 – Task 1

Table Booking & Online Delivery

Objective

Analyze table booking and online delivery availability.

Work Done

- Percentage of Table Booking
- Percentage of Online Delivery
- Average Rating Comparison
- Price Range Analysis

Result

Restaurants with table booking generally have better average ratings.
Online delivery is more common in lower and medium price ranges.

Level 2 – Task 2

Price Range Analysis

Objective

Analyze restaurant ratings across different price ranges.

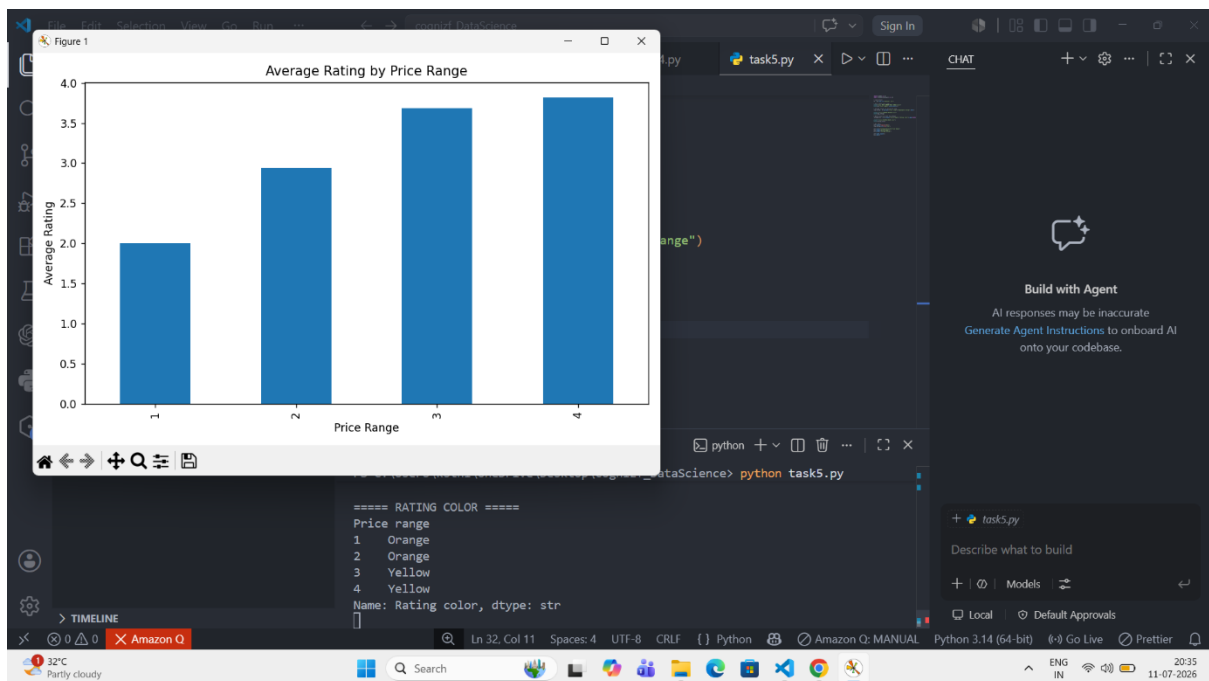
Work Done

- Most Common Price Range
- Average Rating
- Rating Color
- Bar Chart

Result

Higher price ranges generally have higher average ratings.

Screen shot:-



Level 2 – Task 3

Feature Engineering

Objective

Create new features from the existing dataset.

Work Done

- Restaurant Name Length
- Address Length
- Table Booking Encoding
- Online Delivery Encoding

Result

A new feature-engineered dataset was successfully created and saved as **Feature_Engineered_Dataset.csv**.

Skills Learned

- Python Programming
- Pandas
- Data Cleaning
- Data Visualization
- Feature Engineering
- Data Analysis
- VS Code

Conclusion

This internship provided valuable practical experience in Data Science. I successfully completed all assigned Level 1 and Level 2 tasks using Python and data analysis techniques. Through this internship, I improved my skills in data preprocessing, visualization, statistical analysis, and feature engineering.